What is <!DOCTYPE html>

All HTML documents must start with a <!DOCTYPE> declaration.
The declaration is not an HTML tag. It is "information" to the browser about what document type to expect.

In HTML 5, the declaration is simple:

<!DOCTYPE html>

Older HTML Documents

In older documents (HTML 4 or XHTML), the declaration is more complicated because the declaration must refer to a DTD (Document Type Definition).
HTML 4.01:

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
```

XHTML 1.1:

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN" "http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd">
```

What are HTML Attributes?

Attributes are the properties that can be added to an HTML tag. These attributes change the way the tag behaves or is displayed. For example, a <img> tag has an src attribute, which you use to add the source from which the image should be displayed.
We add attributes right after the name of the HTML tag, inside the brackets.
We can only add the attributes to opening or self-closing tags, but never be in closing tags.

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What is white space in HTML?

An empty sequence of space characters is called the white space in HTML. This white space is considered a single space character in the HTML. White space helps the browser to merge multiple spaces into one single space, so taking care of indentation becomes easier. White space helps in better organizing the content and tags, making them readable and easy to understand.

What is DOM (Document Object Model)

With the HTML DOM, JavaScript can access and change all the elements of an HTML document.
The HTML DOM (Document Object Model)
When a web page is loaded, the browser creates a Document Object Model of the page.
The HTML DOM model is constructed as a tree of Objects:
With the object model, JavaScript gets all the power it needs to create dynamic HTML:

- JavaScript can change all the HTML elements on the page
- JavaScript can change all the HTML attributes on the page
- JavaScript can change all the CSS styles on the page
- JavaScript can remove existing HTML elements and attributes
- JavaScript can add new HTML elements and attributes
- JavaScript can react to all existing HTML events on the page
- JavaScript can create new HTML events on the page

The HTML DOM is a standard object model and programming interface for HTML. It defines:

- The HTML elements as objects
- The properties of all HTML elements
- The methods to access all HTML elements

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- The events for all HTML elements

In other words: The HTML DOM is a standard for how to get, change, add, or delete HTML elements

What are Semantic Elements?

A semantic element clearly describes its meaning to both the browser and the developer.
Examples of non-semantic elements: and - Tells nothing about its content.
Examples of semantic elements: <form>, <table>, and <article> - Clearly define its content.

Semantic Elements in HTML

Many websites contain HTML code like: to indicate navigation, header, and footer.

In HTML there are some semantic elements that can be used to define different parts of a web page:

- <article>
- <aside>
- <details>
- <figcaption>
- <figure>
- <footer>
- <header>
- <main>
- <mark>
- <nav>
- <section>
- <summary>
- <time>

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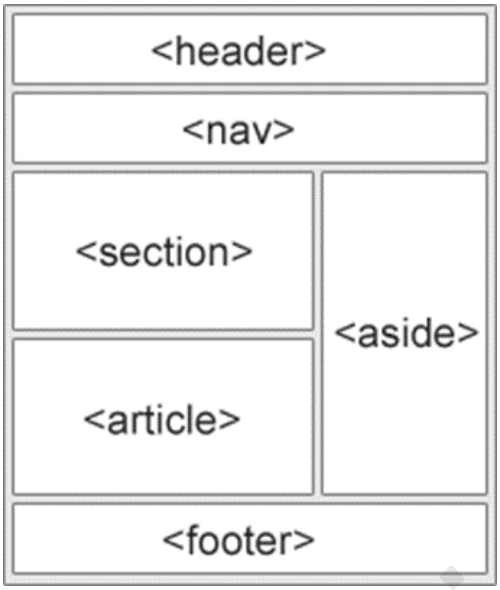

HTML <section> Element

The **<section>** element defines a section in a document.

According to W3C's HTML documentation: "A section is a thematic grouping of content, typically with a heading."

Examples of where an **<section>** element can be used:

- Chapters
- Introduction
- News items
- Contact information

A web page could normally be split into sections for introduction, content, and contact information.

HTML <meta> Tag

```
<head>
  <meta charset="UTF-8">
  <meta name="description" content="Free Web tutorials">
  <meta name="keywords" content="HTML, CSS, JavaScript">
  <meta name="author" content="John Doe">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
```

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The <meta> tag defines metadata about an HTML document. Metadata is data (information) about data.

<meta> tags always go inside the <head> element and are typically used to specify a character set, page description, keywords, author of the document, and viewport settings.

Metadata will not be displayed on the page but is machine parsable.

Metadata is used by browsers (how to display content or reload pages), search engines (keywords), and other web services.

Attributes

Attribute	Value	Description
charset	*character_set*	Specifies the character encoding for the HTML document
content	*text*	Specifies the value associated with the http-equiv or name attribute
http-equiv	content-security-policy content-type default-style refresh	Provides an HTTP header for the information/value of the content attribute
name	application-name author description generator keywords viewport	Specifies a name for the metadata

Local Storage, Session Storage, and Cookies

are all ways to store data on the client side in a web browser. However, there are some differences between them:

Scope

Cookies: Cookies are sent to the server with each HTTP request, so they can be accessed from both the client and server sides. Cookies are also shared between tabs and windows.

Local Storage: Local Storage is only accessible from the domain that created it. Local Storage is not shared between tabs and windows.

Session Storage: Session Storage is similar to Local Storage, but the data is cleared when the browser is closed. Session Storage is not shared between tabs and windows.

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Size

Cookies: Cookies are limited to 4KB of data.

Local Storage: Local Storage can store up to 10MB of data.

Session Storage: Session Storage can store up to 5-10MB of data.

Expiration:

Cookies: Cookies can have an expiration date, after which they will be deleted.

Local Storage: Local Storage does not have an expiration date.

Session Storage: Session Storage is deleted when the browser is closed.

Access:

Cookies: Cookies can be accessed by both the client and server side.

Local Storage: Local Storage can only be accessed by the client side.

Session Storage: Session Storage can only be accessed by the client side.

In summary, Cookies are useful for storing small amounts of data that need to be shared between the client and server side. Local Storage and Session Storage are useful for storing larger amounts of data that are only needed on the client side. Local Storage is more permanent than Session Storage, which is cleared when the browser is closed.

Rapid Fire Interview Questions and answers:

What is HTML? Explain its basic structure.

HTML stands for HyperText Markup Language. It is used to create web pages and web applications. The basic structure of an HTML document consists of an HTML tag, a head tag, and a body tag. The HTML tag identifies the document as an HTML document. The head tag contains information about the document, such as the title and the links to stylesheets and scripts. The body tag contains the content of the document, such as text, images, and videos.

What are the different versions of HTML, and what are the differences between them?

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The different versions of HTML are HTML, HTML 4.01, XHTML 1.0, HTML5, and XHTML5. The main differences between these versions are the syntax, the structure, and the features. HTML5 and XHTML5 are the latest versions and have many new features, such as new elements, multimedia support, and canvas drawing.

What are the new features of HTML5?

Some of the new features of HTML5 are new semantic tags, multimedia support, canvas drawing, geolocation, web storage, and web workers.

What is the purpose of the DOCTYPE declaration in HTML?

The DOCTYPE declaration is used to specify the version of HTML that is being used in the document. It is also used to tell the web browser how to render the page.

What is the difference between HTML and XHTML?

The main difference between HTML and XHTML is the syntax. HTML uses a loose syntax, while XHTML uses a strict syntax that requires all elements to be properly nested and closed.

What is the difference between HTML elements and HTML attributes?

HTML elements are the building blocks of HTML documents. They define the structure and content of the document. HTML attributes provide additional information about an element, such as the color of text or the width of an image.

What are semantic tags in HTML?

Semantic tags in HTML are tags that describe the meaning of the content they contain, rather than just their appearance. Examples of semantic tags are <header>, <nav>, <main>, <article>, <section>, <aside>, and <footer>.

What is the purpose of the <meta> tag in HTML?

The <meta> tag is used to provide metadata about the HTML document, such as the character encoding, keywords, description, and author.

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How do you include images in an HTML page?

Images can be included in an HTML page using the <img> tag, which has a src attribute that specifies the URL of the image file.

What are the different types of lists in HTML?

The different types of lists in HTML are unordered lists (<ul>), ordered lists (<ol>), and definition lists (<dl>).

What is the difference between an inline element and a block element in HTML?

An inline element is an element that can be placed within a line of text, such as <a> or . A block element is an element that creates a new block of content, such as <p> or .

How do you create a hyperlink in HTML?

Hyperlinks can be created in HTML using the <a> tag, which has an href attribute that specifies the URL of the destination.

What is the purpose of the alt attribute in an image tag?

The alt attribute in an image tag provides alternative text for the image, which is displayed when the image cannot be displayed or is being read by a screen reader.

How do you embed a video in an HTML page?

Videos can be embedded in an HTML page using the <video> tag, which has a source attribute that specifies the URL of the video file.

Rapid Fire Interview Question:

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What is HTML? Explain its basic structure.

What are the different versions of HTML, and what are the differences between them?

What are the new features of HTML5?

What is the purpose of the DOCTYPE declaration in HTML?

What is the difference between HTML and XHTML?

What is the difference between HTML elements and HTML attributes?

What are semantic tags in HTML?

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How do you embed a video in an HTML page?

What is the difference between the and tags in HTML?

What is the purpose of the <head> tag in HTML?

What is the difference between the and tags in HTML?

How do you create a table in HTML?

What is the purpose of the <form> tag in HTML?

How do you validate an HTML page?

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